

Course No: CSE 4255
Course Title: Telecommunication

Lecture 1: Introduction to Telecommunication

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Course Objective:

- The objective of the course is to give an introduction to the telecommunications, fundamentals of wireless communication system, the wireless network architectures and protocols. Topics of study includes an overview of wired telephony system; switching technique; wireless telecommunication system; cellular telephony; GSM; signal propagation models; fading in the mobile environment; multiple access techniques; modulation techniques; satellite communication and wireless networking standards IEEE-802.11.

Text Books

- ▶ Wireless Communications and Networks; 2nd edition---William Stallings
- ▶ Wireless Communications Principles and Practice; 2nd edition---T. S. Rappaport

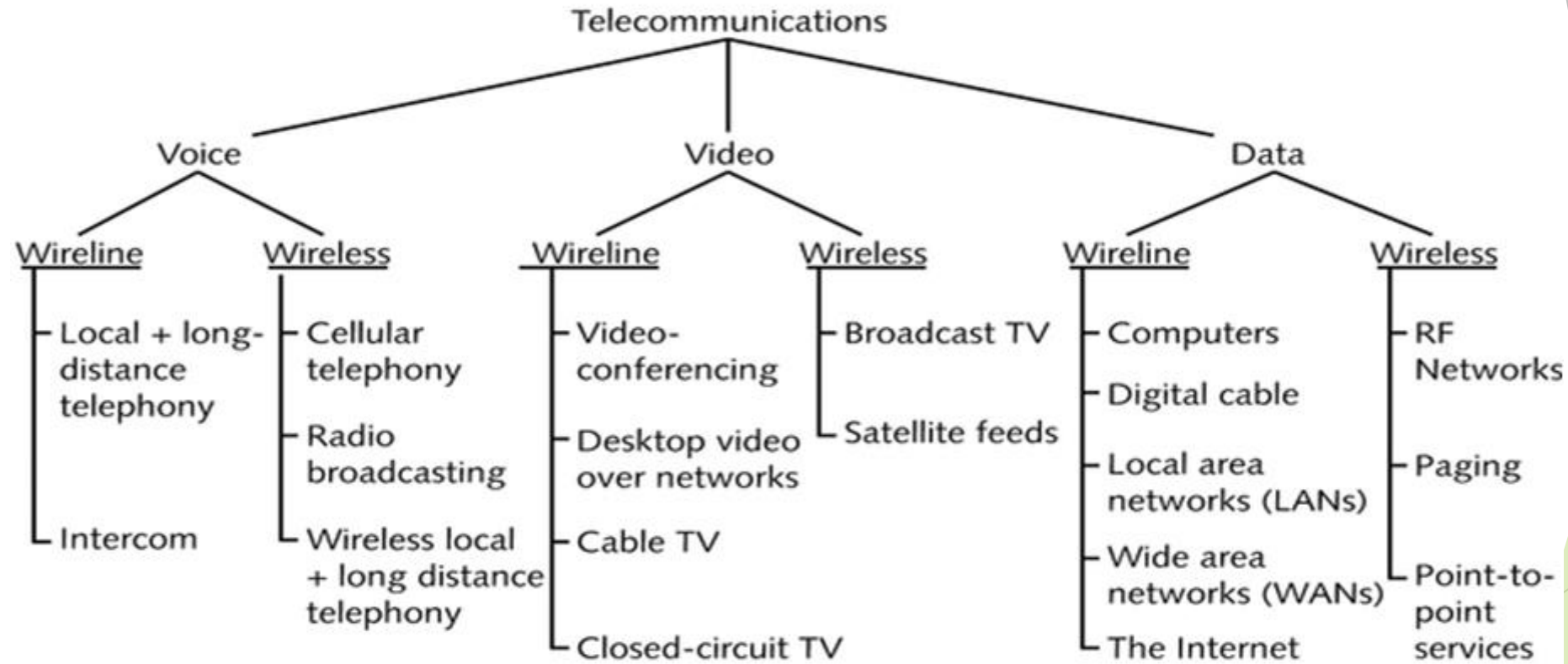
Grading:

- ▶ Class Performance: 10%
- ▶ Quiz: 20%
- ▶ Final: 70%

What is telecommunication?

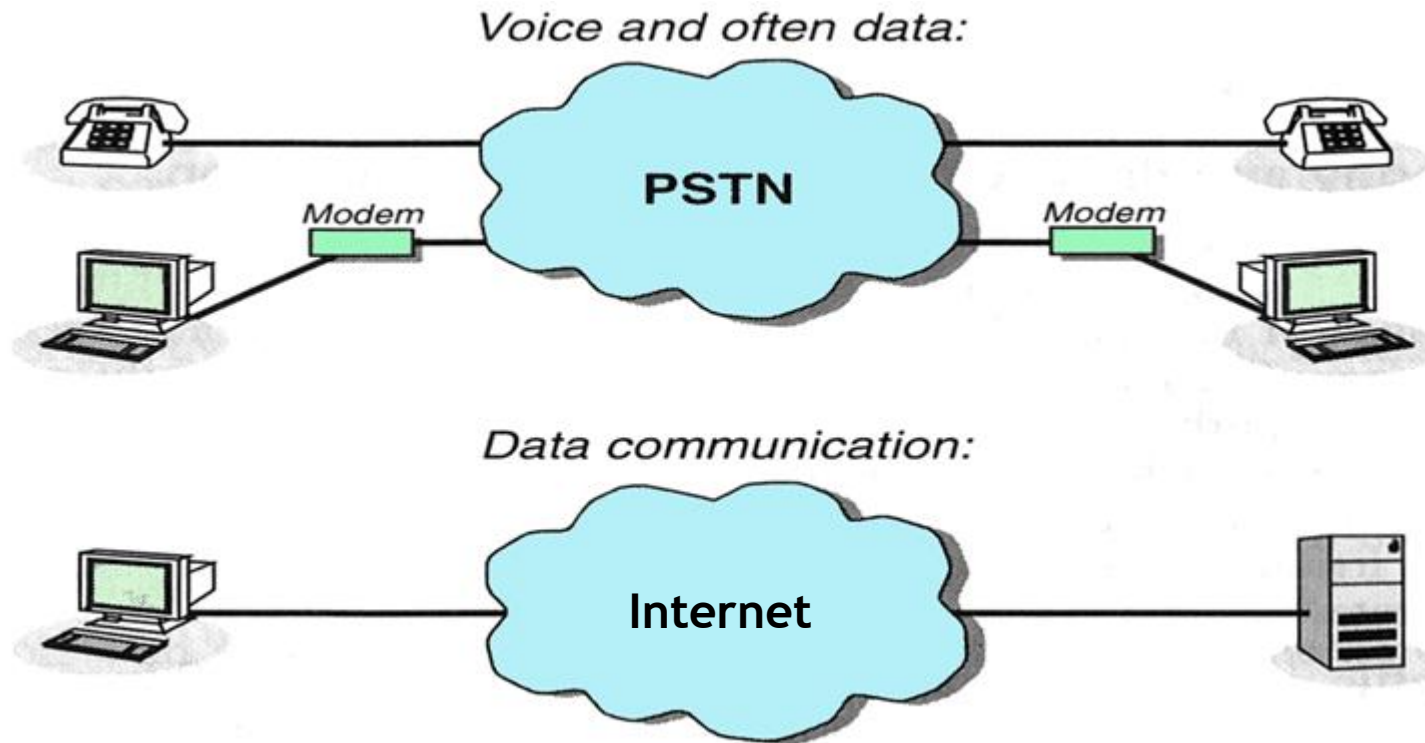
- ▶ **Communication** is the sharing of information or messages between two or more entities.
- ▶ The word ***tele*** is Greek for "far off".
- ▶ The term ***telecommunication***, means communication at a distance.
- ▶ The information may be in the form of voice, video, and data:
 - ▶ **Voice telecommunication** - using electrical signals to transmit human voice across a distance, such as telephones and radio broadcasts.
 - ▶ **Video telecommunication** - the electrically-based transmission of moving pictures and sound across a distance.
 - ▶ **Data telecommunication** - the use of electrical signals to exchange encoded information between computerized devices across a distance.

Types of Telecommunication Services



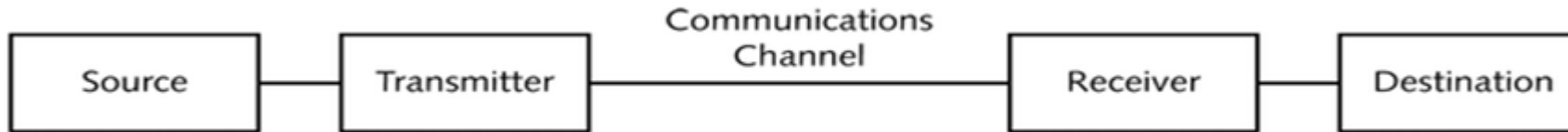
Data and Voice Networks

Divided networks



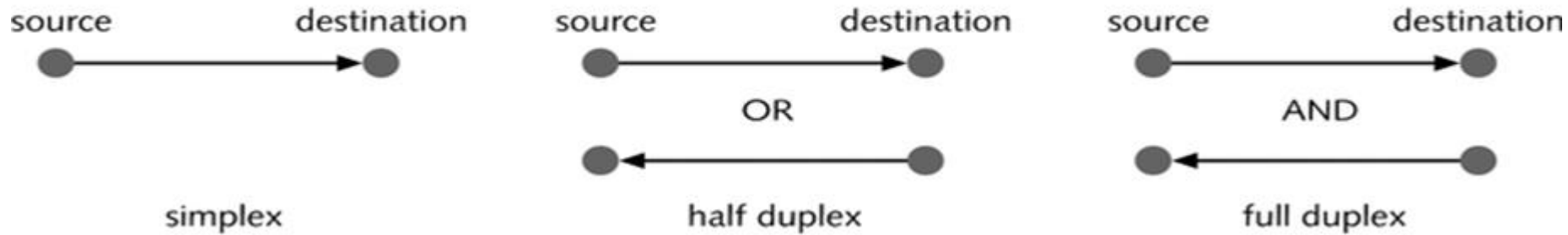
- ▶ Voice telecommunication or Switched Telephone Network - PSTN, ISDN (integrated services digital network), GSM
- ▶ Data telecommunication or Computer (data) networks - Internet, WLAN,

Elements of Telecommunication System



- ▶ **Source** - the originator of the message, whether it is a person or machine.
- ▶ **Transmitter** - the equipment that modifies the message (either data or voice) into the form required for transmission.
- ▶ **Communications channel** - the means of carrying the signal from the source to the destination.
- ▶ **Transmission media** - may be physical, like a copper wire or fiber optic cable, or atmospheric, like radio waves.
- ▶ **Receiver** - is the device that captures the message from the communications channel and converts it into a form that the person or machine at the destination can understand.
- ▶ **Destination** - the person or machine to whom the message is directed

Transmission Modes

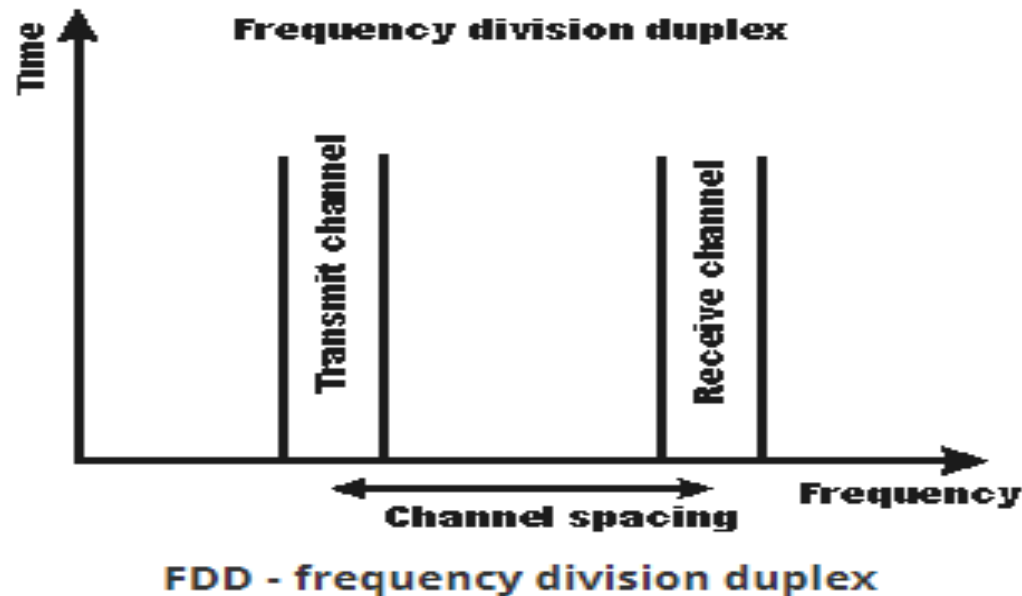


Signals can be transmitted through telecommunications media in a number of different ways:

- ▶ In ***simplex transmission***, data can flow only in one direction
- ▶ In ***half-duplex transmission***, data can flow in both directions but it can only flow in one direction at any point in time
- ▶ In ***full-duplex transmission***, data can flow in both directions at the same time
 - ▶ Frequency-division duplex (FDD) ,,
 - ▶ Time-division duplex (TDD): simulated full-duplex

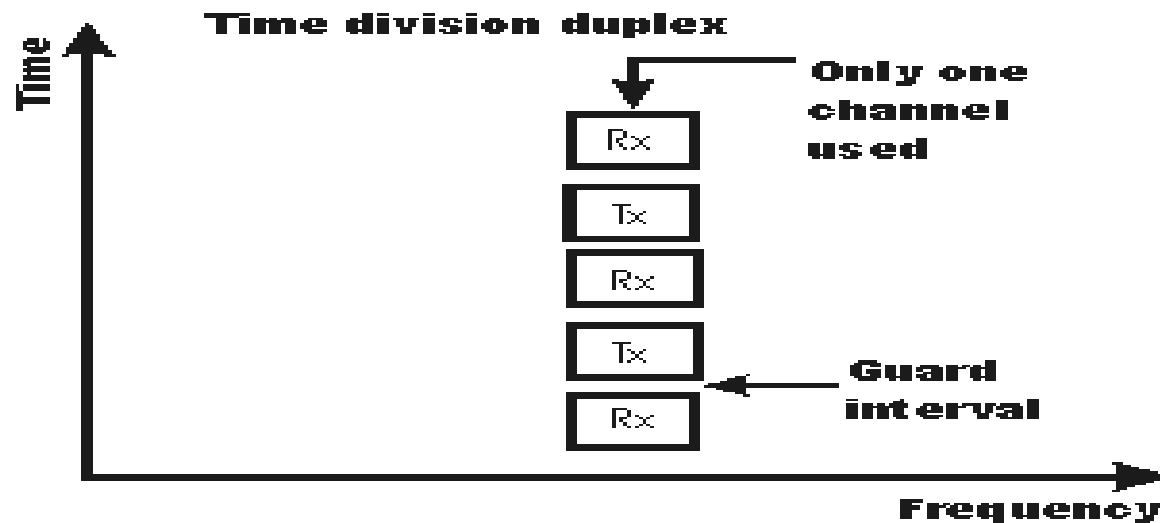
FDD

- Frequency-division duplexing (FDD) is a method for establishing a full-duplex communications link that uses two different radio frequencies for transmitter and receiver operation.
- The transmit direction and receive direction frequencies are separated by a defined frequency offset (guard band).



TDD

- ▶ Time-division duplexing (TDD) is a method for emulating full-duplex communication over a half-duplex communication link.
- ▶ TDD divides a data stream into frames and assigns different time slots to forward and reverse transmissions, thereby allowing both types of transmissions to share the same transmission medium.
- ▶ TDD schemes require a guard time or guard interval between transmission and reception.



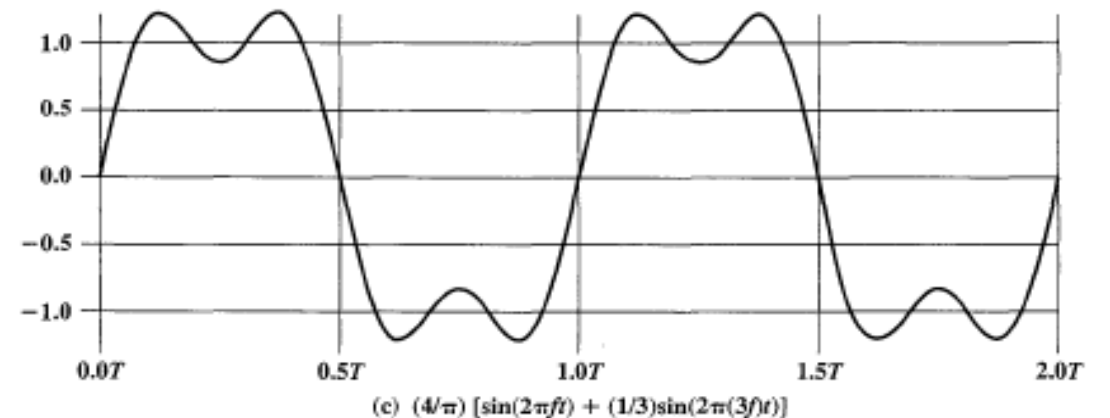
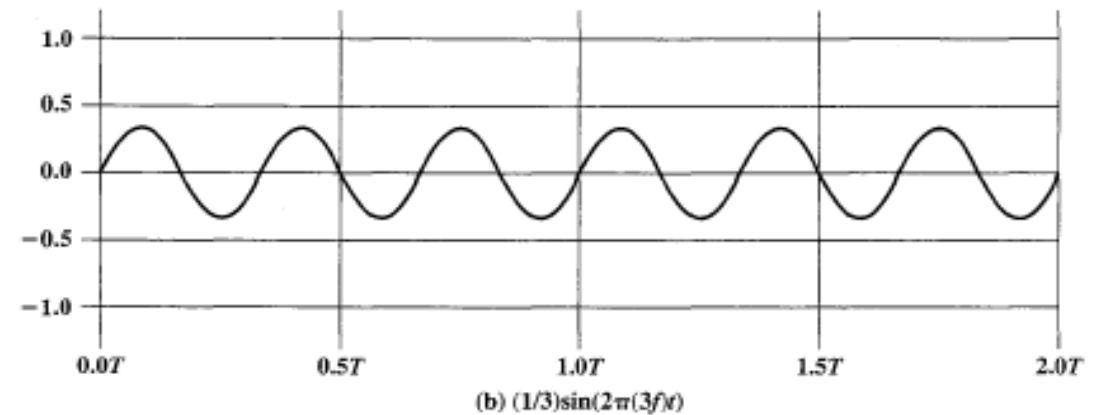
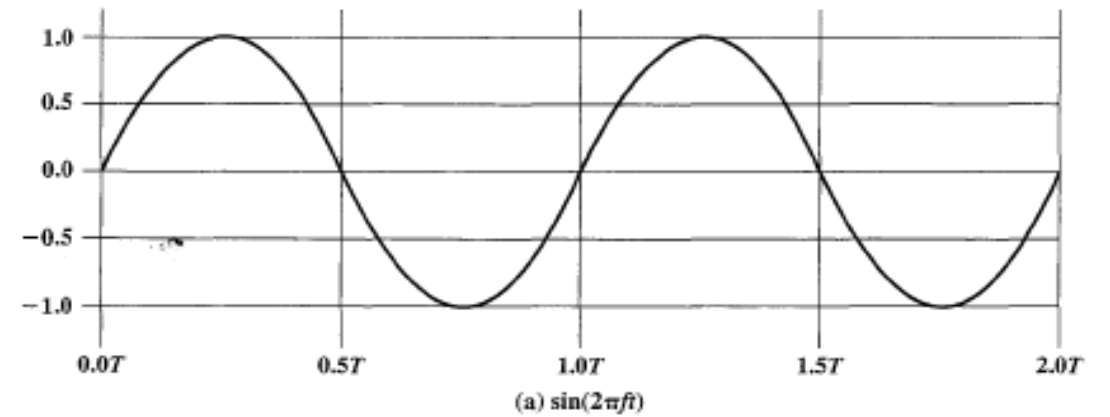
Related Terminologies

- ▶ **Bandwidth** in hertz, refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.
- ▶ **Data Rate** in bits per second, refers to the number of bits per second that a channel, a link, or even a network can transmit.
- ▶ **Throughput** is the actual amount of data that is successfully sent/received over the communication link.
- ▶ **Latency (Delay)** defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.
- ▶ **Latency = propagation time + transmission time + queuing time + processing delay.**

Data Rate and Bandwidth

- ▶ Suppose:
 - ▶ Positive pulse → binary 0
 - ▶ Negative pulse → binary 1
- ▶ Duration of each pulse is $1/2f$
- ▶ Data rate is $2f$ bps
- ▶ If more frequencies are added to the signal then the wave form approaches to the square wave
- ▶ The frequency components of the square wave with amplitude A and $-A$ can be expressed as follows :

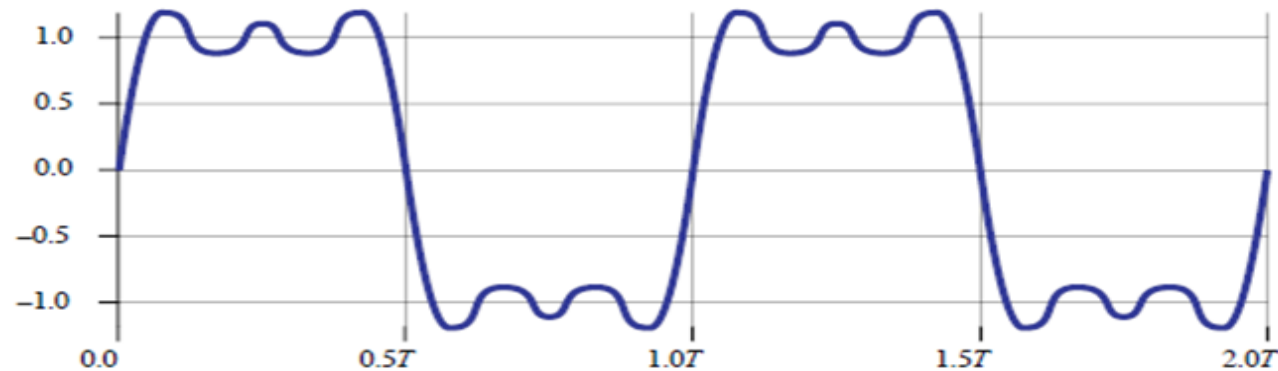
$$s(t) = A \times (4/\pi) \times \sum_{k \text{ odd}, k=1}^{\infty} \sin(2\pi f t)/k$$



Data Rate Calculation

► Case 1: Figure a

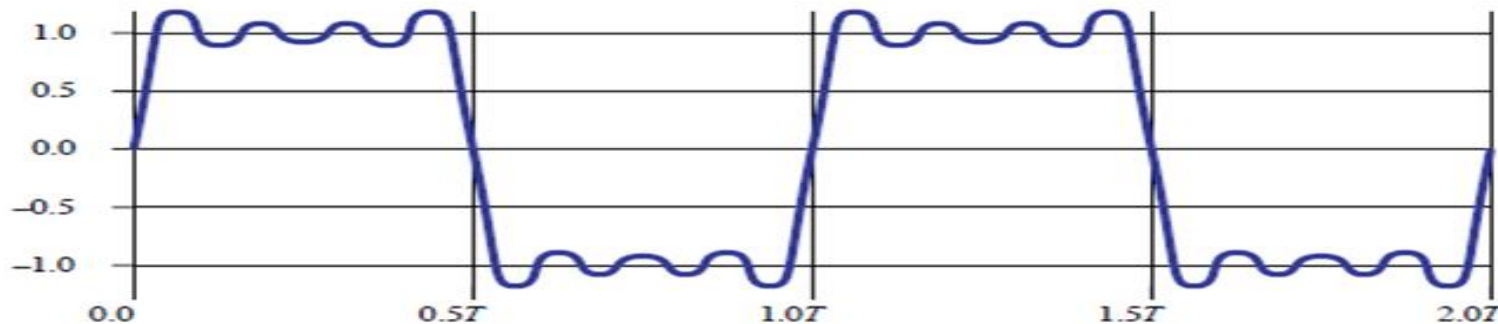
- Let $f = 10^6$ cycles/sec = 1 MHz
- Frequency components = $1f, 3f, 5f$
- Bandwidth = $5f - f = 4f = 4$ MHz
- $T = 1/10^6 = 10^{-6} = 1\mu\text{s}$
- If we treat this wave form as bit string of 1s and 0s, 1 bit occurs at every $0.5\mu\text{s}$, i.e duration of each pulse is $1/2 * 10^6$
- Data rate = $2 * 10^6 = 2\text{Mbps}$
- Thus for 4 MHz BW data rate is 2 Mbps



(a) $(4/\pi) [\sin(2\pi ft) + (1/3) \sin(2\pi(3f)t) + (1/5) \sin(2\pi(5f)t)]$

Data Rate Calculation

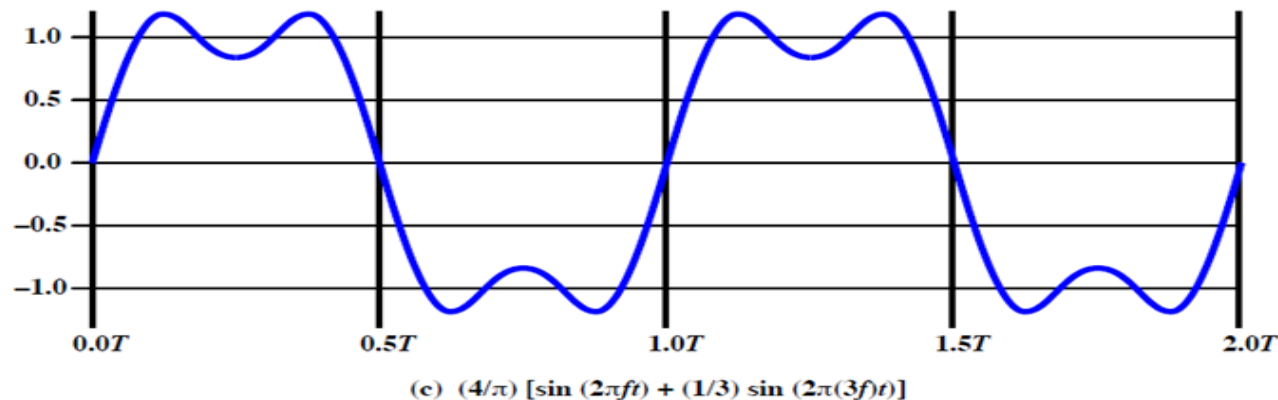
- ▶ Case 2: Figure b
 - ▶ Let $f = 2 \times 10^6$ cycles/sec = 2 MHz
 - ▶ Frequency components = $1f, 3f, 5f$
 - ▶ Bandwidth = $5f - f = 4f = (5 \times 2 \times 10^6 - 2 \times 10^6) = 8$ MHz
 - ▶ $T = 1 / 2 \times 10^6 = 0.5 \mu\text{s}$
 - ▶ If we treat this wave form as bit string of 1s and 0s, 1 bit occurs at every $0.25 \mu\text{s}$ i.e duration of each pulse is $1/4 \times 10^6$
 - ▶ Data rate = $4 \times 10^6 = 4$ Mbps
 - ▶ Thus for 8 MHz BW data rate is 4 Mbps



(b) $(4/\pi) [\sin(2\pi ft) + (1/3) \sin(2\pi(3f)t) + (1/5) \sin(2\pi(5f)t) + (1/7) \sin(2\pi(7f)t)]$

Data Rate Calculation

- ▶ Case 3: Figure c
 - ▶ Let $f = 2 \times 10^6$ cycles/sec = 2 MHz
 - ▶ Frequency components = $1f, 3f$
 - ▶ Bandwidth = $3f - f = 2f = (3 \times 2 \times 10^6 - 2 \times 10^6) = 4\text{MHz}$
 - ▶ $T = 1 / 2 \times 10^6 = 0.5 \mu\text{s}$
 - ▶ If we treat this wave form as bit string of 1s and 0s, 1 bit occurs at every $0.25 \mu\text{s}$ i.e. duration of pulse is $1/4 \times 10^6$
 - ▶ Data rate = $4 \times 10^6 = 4\text{Mbps}$
 - ▶ Thus for 4 MHz BW data rate is 4 Mbps



Data Rate Vs Bandwidth

- ▶ Bandwidth ↑
 - ▶ Data rate ↑ (compare case 1 and 2)
 - ▶ Same signal quality
- ▶ Same bandwidth
 - ▶ Higher signal quality → lower data rate
 - ▶ Compare case 1 and 3
- ▶ Same data rate
 - ▶ Bandwidth ↑ → better signal quality
 - ▶ Compare case 2 and 3

To summarize,

- **Case I:** Bandwidth = 4 MHz; data rate = 2 Mbps
- **Case II:** Bandwidth = 8 MHz; data rate = 4 Mbps
- **Case III:** Bandwidth = 4 MHz; data rate = 4 Mbps

Data Rate Vs Bandwidth

- ▶ Same signal quality (Compare case 1 and 2)
 - ▶ Bandwidth $\uparrow$
 - ▶ Data rate $\uparrow$
- ▶ Same bandwidth (Compare case 1 and 3)
 - ▶ Higher signal quality $\rightarrow$ Lower data rate
 - ▶ Lower signal quality $\rightarrow$ Higher data rate
- ▶ Same data rate (Compare case 2 and 3)
 - ▶ Bandwidth $\uparrow$
 - ▶ Higher signal quality

To summarize,

- **Case I:** Bandwidth = 4 MHz; data rate = 2 Mbps
- **Case II:** Bandwidth = 8 MHz; data rate = 4 Mbps
- **Case III:** Bandwidth = 4 MHz; data rate = 4 Mbps

Channel Capacity

- ▶ The maximum rate at which data can be transmitted over a given communication path, or channel, under given conditions.
- ▶ Impairments, such as noise, limit data rate that can be achieved.
- ▶ Nyquist Formula is: $C = 2B \log_2 M$
- ▶ Shannon Capacity Formula is: $C = B \log_2(1 + \text{SNR})$

Signal to Noise Ratio

- ▶ Signal to noise ratio (SNR) = power of signal/power of noise
 - ▶ Typically measured at the receiver
 - ▶ $\text{SNR}_{\text{db}} = 10 \log_{10}(\text{signal/noise})$
 - ▶ A high SNR means a high quality of signal

Example 1:

- Let's consider the spectrum of a channel is between 3 MHz and 4 MHz and $\text{SNR}_{\text{db}} = 24$ dB. Then find the maximum capacity achieved by channel. If it is possible to achieve the limit then how many signal level is required?

Example 2:

- What are the propagation time and the transmission time for a 2.5-kbyte message if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s.